



Customer Shopping Behavior Analysis

A comprehensive analysis of 3,900 purchases — uncovering spending patterns, customer segments, and product preferences to drive executive decision-making.

Project Overview

Scale

Analyzed **3,900 purchases** across multiple product categories

Goal

Uncover insights into spending patterns, customer segments, and product preferences

Data Scope

Customer demographics, purchase details, and shopping behavior metrics



Dataset Summary

Dataset Structure

3,900 rows with 18 columns capturing customer and transaction data.

Key features: Demographics, purchase details, and shopping behavior metrics.

Missing data: 37 values in the Review Rating column, imputed using median by category to preserve analytical integrity.

3,900

Total transaction records

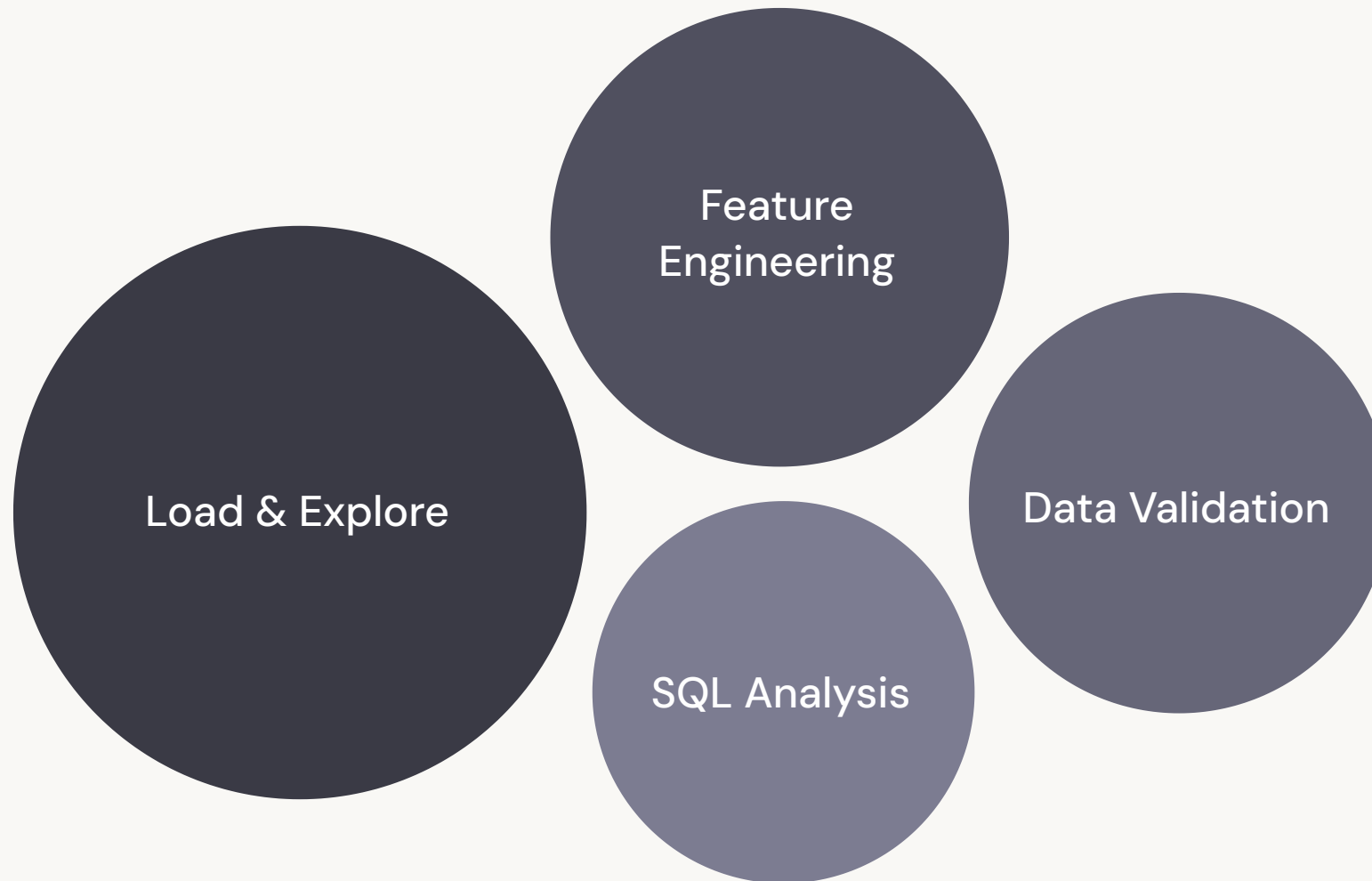
18

Data columns per record

37

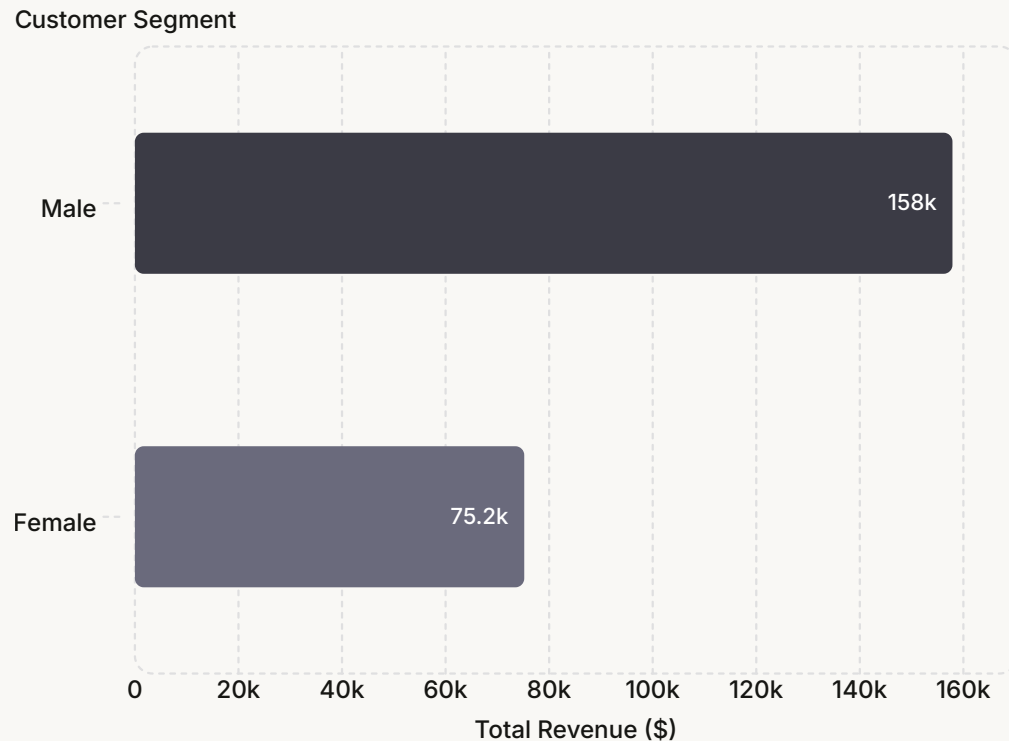
Missing values — imputed

Data Preparation & Cleaning



A rigorous preparation pipeline ensured clean, analysis-ready data — from initial exploration in Python through to structured querying in PostgreSQL.

Revenue by Gender

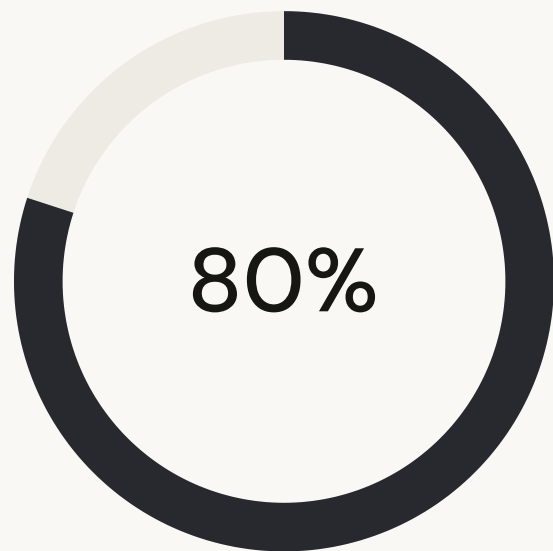


Key Insight

Male customers generate **\$157,890** in total revenue — more than **2.1x** the revenue contributed by female customers at **\$75,191**.

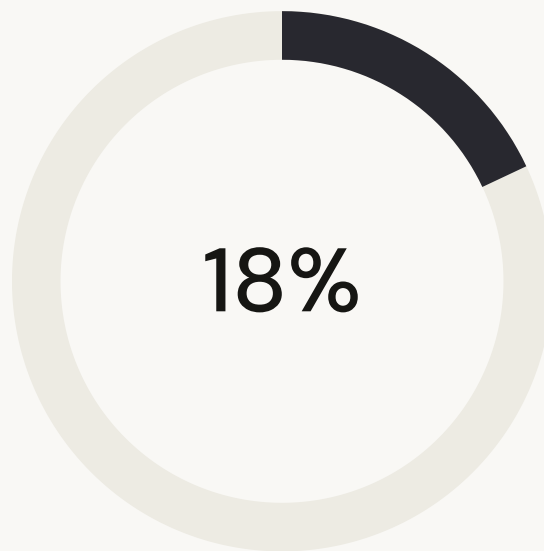
This significant gap signals a clear opportunity to develop targeted campaigns that deepen engagement with female shoppers and unlock untapped revenue potential.

Customer Segmentation



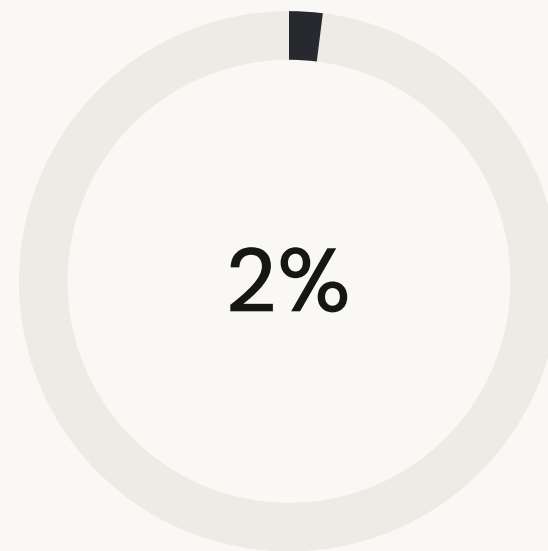
Loyal

3,116 customers — the backbone of revenue



Returning

701 customers — high conversion potential



New

83 customers — early-stage acquisition

The customer base is overwhelmingly loyal, with **80% of customers** classified as repeat buyers. Returning customers represent a strategic opportunity to accelerate movement into the loyal tier.

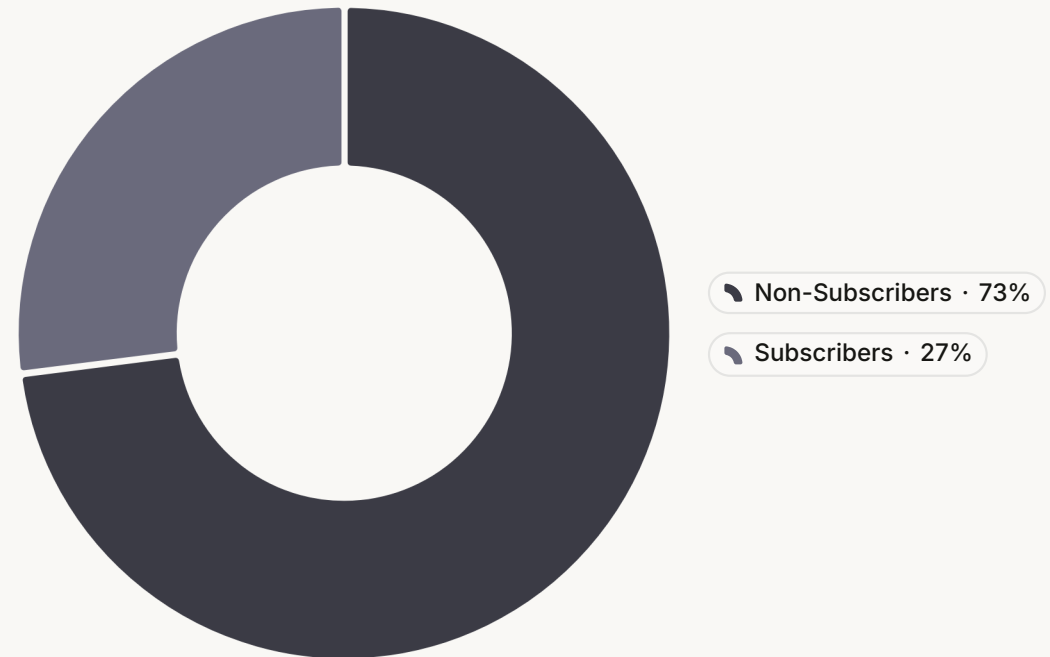
Subscription Insights

Subscription vs. Non-Subscription

Non-subscribers dominate both headcount and revenue — but subscribers punch above their weight on engagement.

- **Subscribers:** 1,053 customers (27%) | \$62,645 total revenue
- **Non-subscribers:** 2,847 customers (73%) | \$170,436 total revenue

Among repeat buyers with more than 5 purchases: **958 subscribers** vs. **2,518 non-subscribers** — indicating strong loyalty behavior exists outside the subscription model.



Top Products by Rating & Sales

Highest-Rated Products

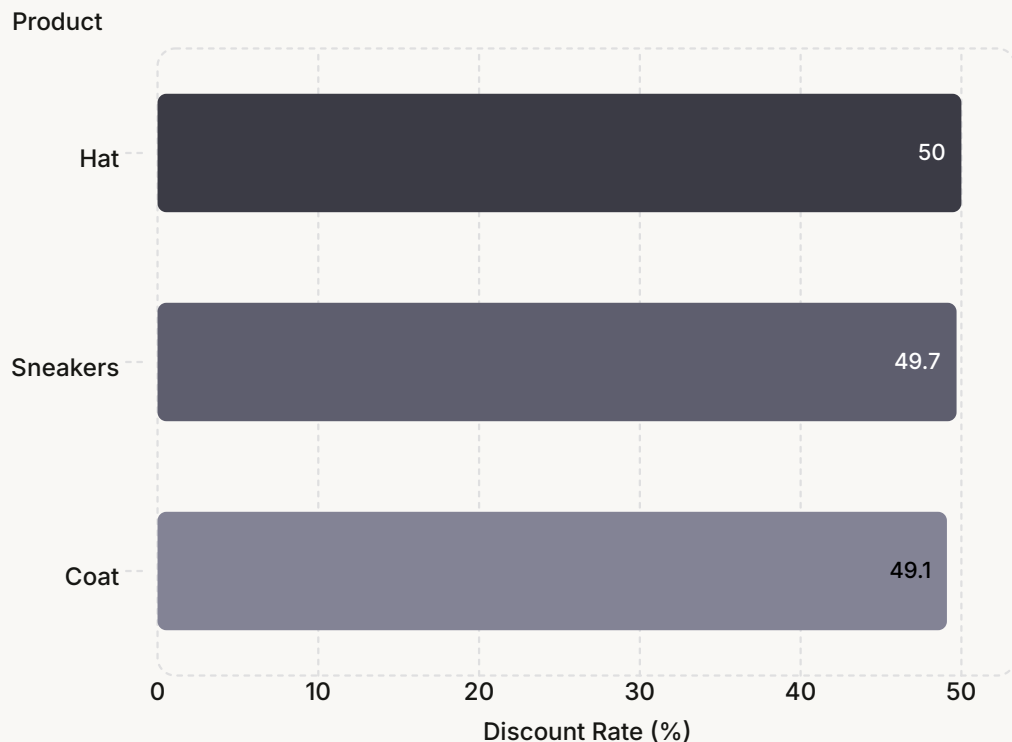
- Gloves — 3.86 ★
- Sandals — 3.84 ★
- Boots — 3.82 ★

Best-Sellers by Category

- Jewelry
- Accessories
- Blouse
- Clothing
- Sandals
- Footwear
- Jacket
- Outerwear



Discount Dependency



The Discount Paradox

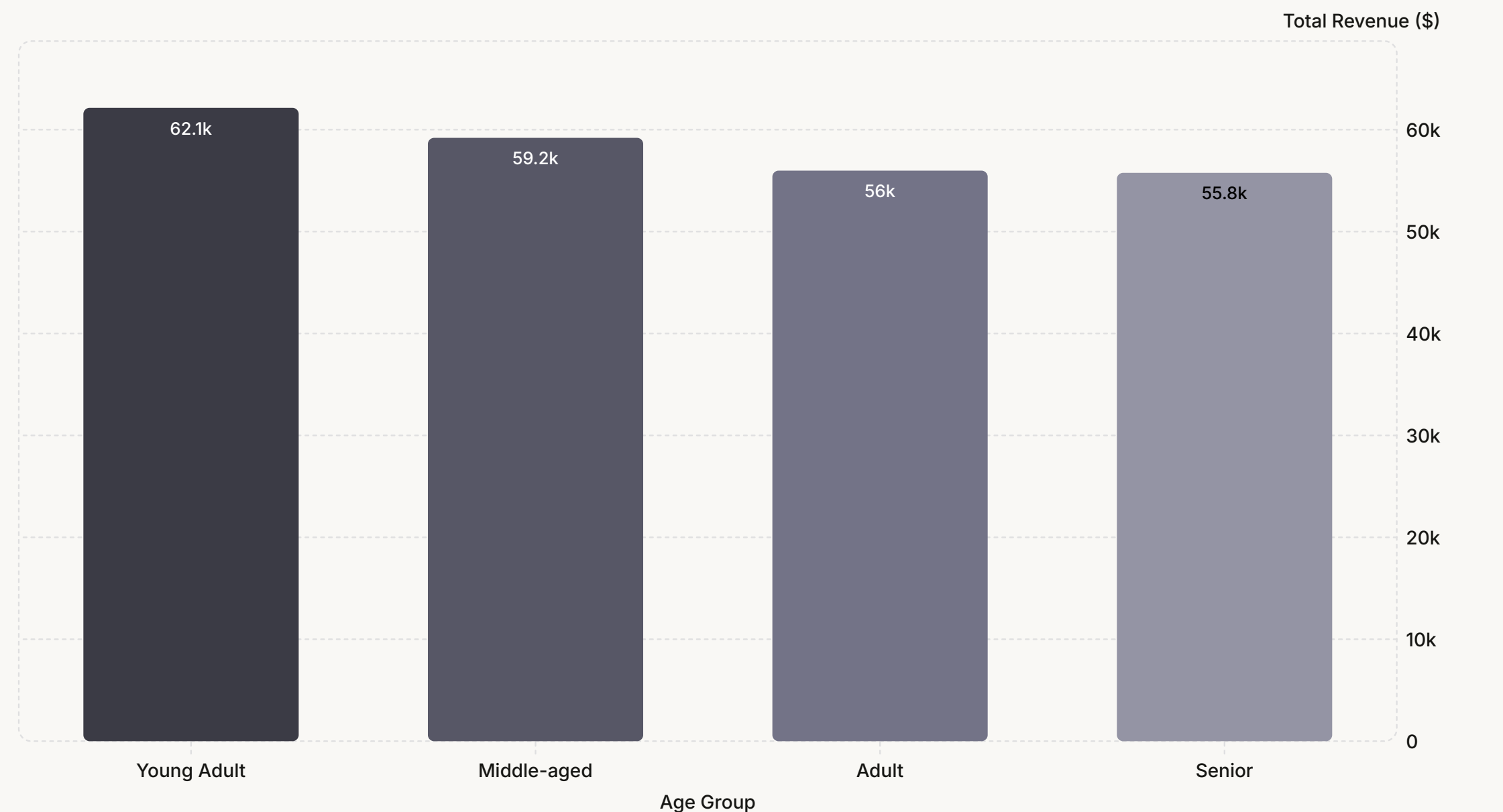
The three highest-discounted products — **Hat (50%)**, **Sneakers (49.7%)**, and **Coat (49.1%)** — carry near-maximum discount rates.

Notably, **839 customers** used discounts yet still spent **above the average purchase amount**, suggesting discounts are driving volume without necessarily sacrificing basket size.



A careful balance between sales boosts and margin control is essential to protect long-term profitability.

Revenue by Age Group



Revenue is remarkably balanced across all age groups, with **Young Adults leading at \$62,143**. The narrow spread — just \$6,380 between the top and bottom segments — indicates broad, cross-generational appeal and a stable, diversified customer base.

Business Recommendations



Boost Subscriptions

Promote exclusive benefits for subscribers to grow the 27% subscriber base



Loyalty Programs

Reward repeat buyers to accelerate movement into the "Loyal" segment



Review Discount Policy

Balance sales boosts with margin control — especially for Hat, Sneakers, and Coat



Targeted Marketing

Focus on high-revenue age groups and express-shipping users for maximum ROI

